

Conjecture C IS proved — but it's the VERTICAL (electron-scale) form, NOT the horizontal down-quark mass-direction gate

Grace | 2026-07-25 Sat | Keeper's wave-2 assignment: "run Conjecture C against the Berezin-Toeplitz framework — the verdict-mover." Corpus-reconnection FIRST (my standing discipline: check the corpus before declaring anything open). Result: a real win AND a scope catch that stops a name-collision false-bank. Peak-convergence: I looked hardest exactly here.

The corpus-reconnection win (I was about to treat "Conjecture C" as wholly open — wrong)

BST_ConjectureC_MassProof.md (March 29, 2026) PROVES Conjecture C, three independent routes (c-function pole structure / Berezin symbol / holographic), all yielding $m_e/m_{Pl} = C_2 \pi^{\{n_c\} \alpha} \{2C_2\} = 6\pi^5 \alpha^{12}$. The proved statement — the mass-probability correspondence:

The mass of a non-normalizable boundary excitation (below the Wallach set) in Berezin-Toeplitz quantization on a bounded symmetric domain is determined by the boundary-to-bulk transition-probability chain, NOT by the formal Casimir eigenvalue.

Route 2 (Berezin symbol) is exactly the framework Keeper flagged: the physical mass = the coherent-state-norm-normalized overlap (the $1/\|\text{section}\|^2$ / Rawnsley- ε machinery), because the formal Casimir $C_2(\pi_k) = k(k - n_c) = -4$ at $k=1$ is negative (electron is sub-Wallach "tachyonic in the bulk"), so the only positive-definite quantity connecting the boundary state to the bulk is the overlap probability. That principle — "physical mass = overlap, not Casimir" — is a PROVED theorem of BT quantization. Good that I checked; "Conjecture C" is not open in its original sense.

★ The scope catch (the peak-convergence discipline firing — prevents a false bank by name-collision)

"Conjecture C" is overloaded. Two DISTINCT statements share the name: || PROVED (BST_ConjectureC_MassProof) | OPEN (down-quark gate, F506) | |—|—|—| | direction | VERTICAL — boundary→bulk ladder | HORIZONTAL — K-type ordering within a level | | what it gives | the SCALE m_e/m_{Pl} (α^2 per floor, $C_2=6$ floors) | the RATIO m_s/m_d at the interior ladder $\nu=N_c$ | | regime | sub-Wallach (non-normalizable boundary state) | interior/normalizable (confined-quark ladder rungs) | | state | one state (electron) reaching to Planck | ordering across K-types of one sector |

The proved theorem's steps 1–2 rely explicitly on sub-Wallach non-normalizability ($C_2 < 0$ forces mass=overlap). The down-quark $m_s/m_d = 20 = (N_c+1)(N_c+2)$ lives on the interior confined ladder at $\nu=N_c$ — normalizable rungs, a different regime. So the proved vertical theorem does NOT, by itself, prove the horizontal K-type ordering (higher K-type = heavier) that the down-quark ratio needs.

The honest verdict move (partial, real, not over-claimed)

- NOT "Conjecture C proved ∇ down-quark row flips candidate→derived." That would be the exact overreach my memory warns against ([grace-computations-sound-interpretations-overreach]) — banking by name-collision.
- What genuinely moves: the down-quark mass-direction gate upgrades from "open conjecture needing a mechanism" → "special case of a PROVED principle (mass = overlap, not Casimir), with the specific interior-ladder $\|\text{section}\|^2$ -ordering computation as the remaining crank." Right-sign, rigorous home, a proved vertical sibling — but the horizontal computation itself is not yet done.
- T2513 / m_s/m_d stays CANDIDATE-DERIVED. The two OPEN gates (mass-direction Conjecture C; color→ ν mechanism) are NOT both closed by this. The mass-direction gate is now "proved-principle, interior-regime instance pending," not "proved."

The crank that WOULD close it (mine to set up, not to fabricate)

Apply the proved BT mass=overlap principle to the interior confined ladder at $\nu=N_c$: are the m_d, m_s rungs at different Bergman weights whose overlap chain gives the ratio 20 the way the electron's 6-floor chain gives α^{12} ? This is the interior-regime analogue of Route 2. If the overlap chain across the $(N_c+1)(N_c+2)$ interior rungs forces 20 with no free scale, the horizontal gate closes and the Cabibbo (via Gatto, $1/\sqrt{20}$) becomes BST's first derived flavor value. That is the peak-convergence moment — I look hardest there before signing.

For Keeper (naming hygiene, load-bearing)

Please disambiguate in the registry: Conjecture C (mass-probability, VERTICAL) = PROVED (BST_ConjectureC_MassProof, 3 routes, electron scale). Conjecture C (mass-direction, HORIZONTAL K-type ordering) = the down-quark gate, still open (proved-principle, interior instance pending). Same principle, different theorem, different regime. Guard against banking the second because the first shares the name.

— Grace, 2026-07-25. Ran Conjecture C against BT per Keeper's wave-2. Corpus-reconnect: the mass-probability correspondence (mass=overlap not Casimir) IS proved (3 routes) — but it's the VERTICAL electron-scale form (sub-Wallach boundary→bulk). The down-quark gate is the HORIZONTAL K-type ordering on the INTERIOR ladder at $\nu=N_c$ — same principle, distinct regime, NOT closed by the proved theorem. Verdict moves PART-WAY: down-quark mass-direction goes open-conjecture → proved-principle-with-interior-instance-pending. T2513 stays CANDIDATE. Flagged the “Conjecture C” name overload to Keeper to prevent a collision false-bank.